

Physics-Informed Neural Networks for a WNT–RA–HOX Model of Colorectal Cancer Stemness

From a nondimensionalized 7-ODE model to sparse-data forward solves, Bayesian identifiability, and neural-mechanistic hybrids.

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01 The model

Seven coupled ODEs for the WNT–RA–HOX axis, nondimensionalized to **36 parameters** across four disease regimes.

02 The PINN

A Fourier-feature network solves the system forward from **40 observations**, then inverts it for the 36 parameters.

03 Bayesian inference

HMC turns the recovery ceiling into a diagnosis: which parameters the data can **never** pin down, and why.

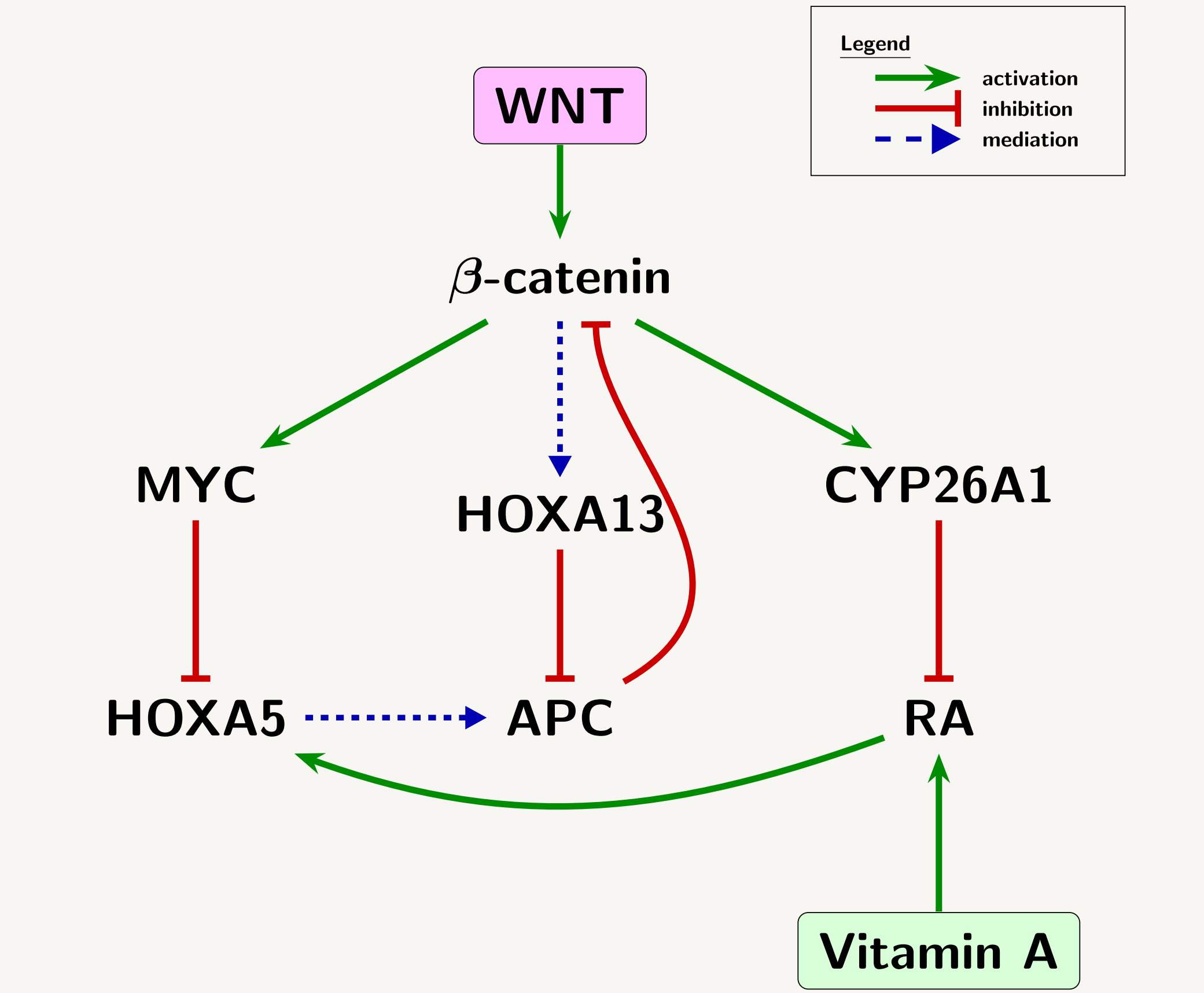
04 Neural-mechanistic hybrid

Replace one mechanistic term with a neural network — and the identifiability problem returns in a new form.

1 THE MODEL

Seven equations, nondimensionalized to 36 parameters.

In the intestinal crypt, WNT signalling drives proliferation, retinoic acid drives differentiation, and the HOX genes arbitrate between them. Loss of APC — the canonical adenoma-to-carcinoma driver — breaks that balance.



Green activates, red inhibits, dashed mediates. Seven species, one constitutive WNT drive, and a vitamin-A input.

Nondimensionalization

01 Scale each species by its characteristic value

taken to be its initial condition, and scale time by the reference β -catenin degradation rate $d_{\beta} = 1 \text{ hr}^{-1}$:
 $B = B_0 b, \quad P = P_0 p, \quad \dots, \quad \tau = d_B t$

β -catenin is WNT's direct target and drives most other species, so its turnover sets the network's reference timescale.

02 Every equation collapses to one form

each loss term normalizes to $-x$, and the prefactor is a **pure timescale ratio**:

$$\varepsilon_X \frac{dx}{d\tau} = \text{production} - x$$

03 The parameter count falls to 36

Nondimensionalization changed the units and the count, not the model. The system is **stiff**: $\varepsilon_R = 0.40, \varepsilon_M = 0.60$.

The scientific readout — the stemness index

$$S(\tau) = \frac{b(1+\alpha_{13}h_{13})}{(1+p)(1+\alpha_5h_5)}$$

Large when proliferative species dominate, small when differentiating species do. Not a state variable — the aggregate the model exists to predict.

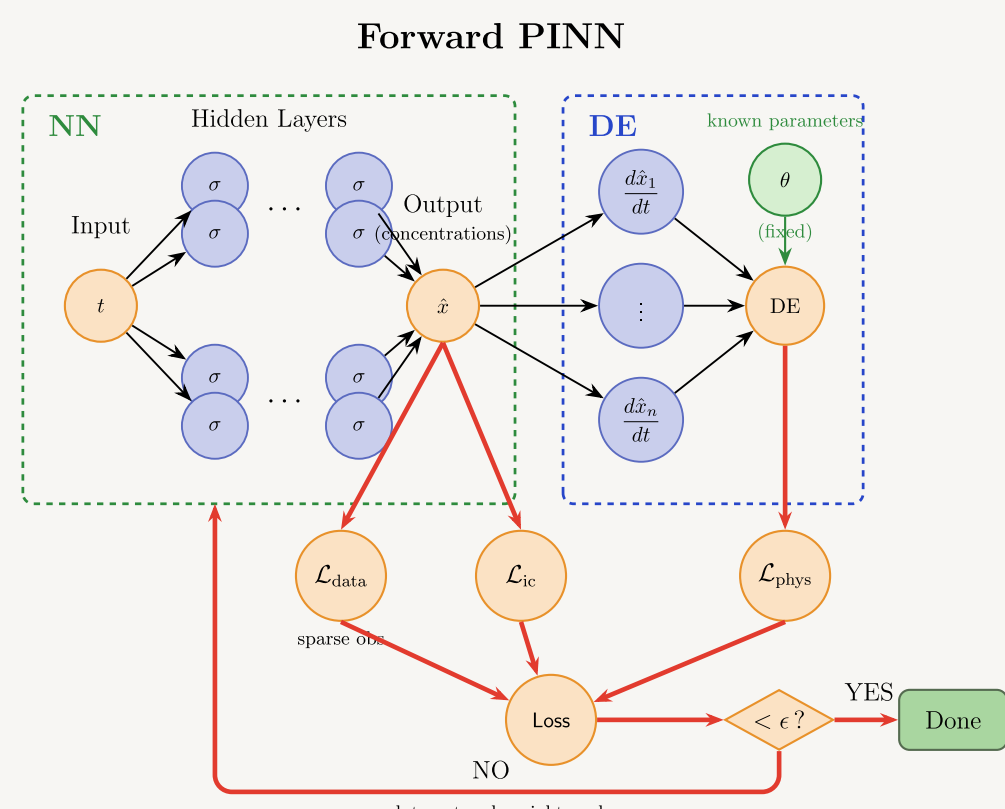
Four regimes, two parameters

Normal $\rightarrow$ Early Adenoma $\rightarrow$ Advanced Adenoma $\rightarrow$ Severe APC Loss differ **only** in the WNT drive W ($0.8 \rightarrow 2.0$) and APC functionality θ_P ($1.0 \rightarrow 0.25$).

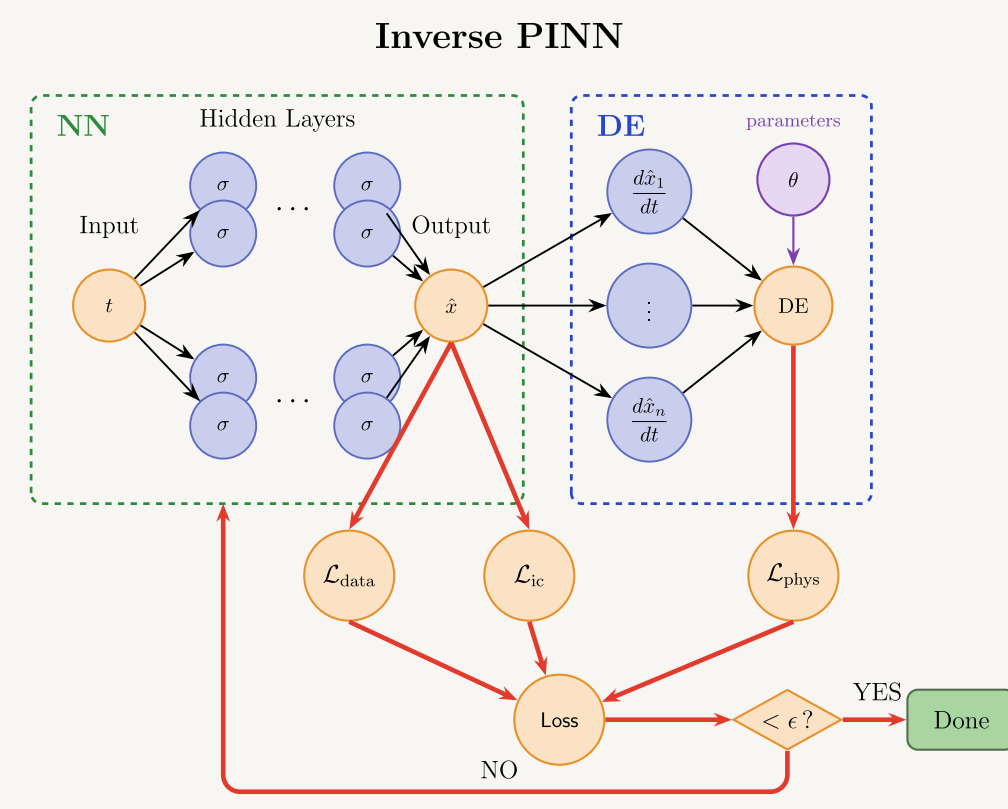
2 THE PINN

One network solves it from 40 points — and inverts it.

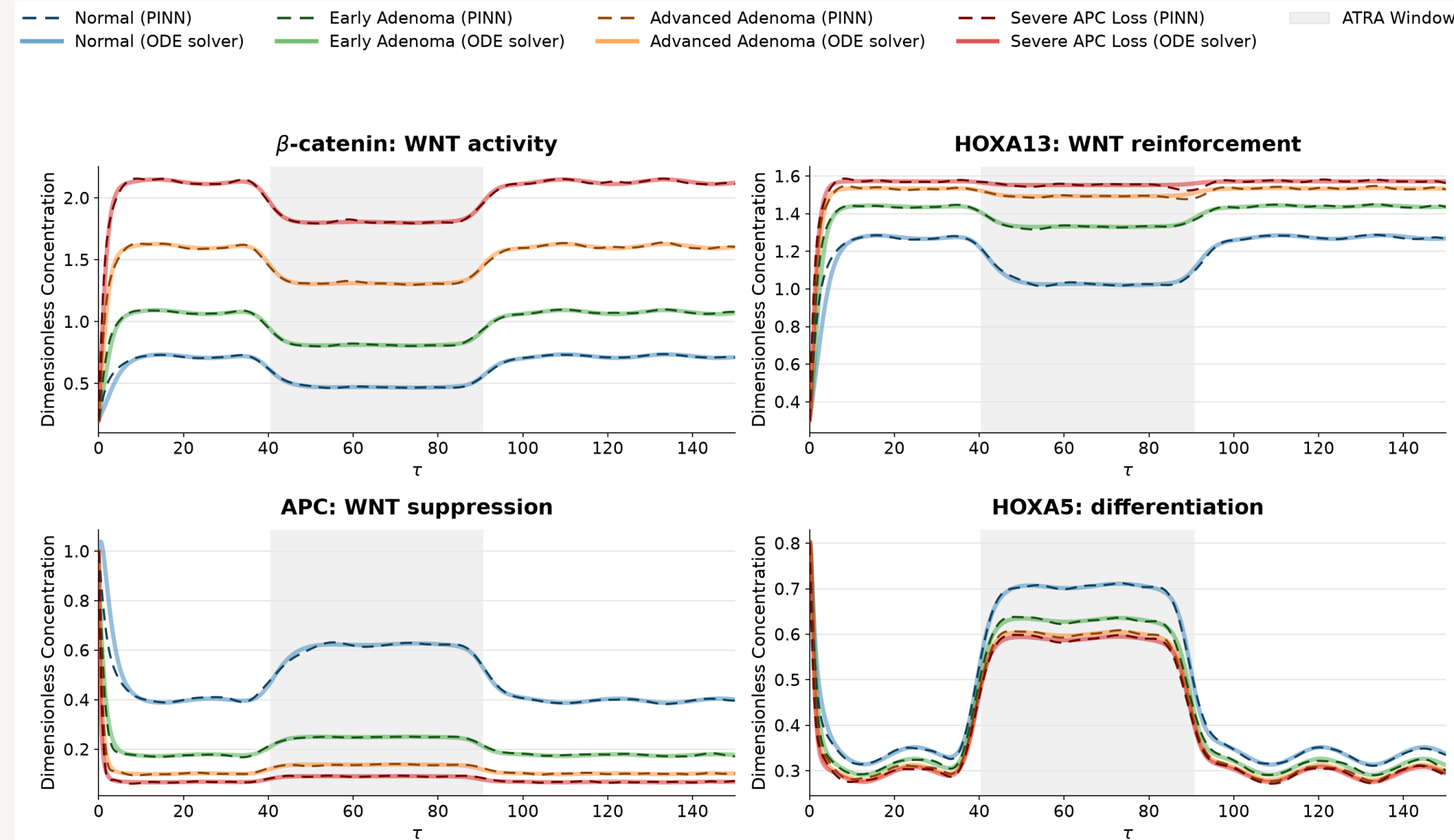
FORWARD — θ fixed



INVERSE — θ trainable



The network proposes a trajectory; automatic differentiation feeds it back through the ODEs. Three losses — data, initial condition, physics. Freeing θ turns the solver into a parameter estimator. A Fourier-feature time embedding (16 modes, $\sigma = 4$) into a 256-wide, 4-deep GELU network — 207,879 weights. Without it, spectral bias renders the circadian forcing as straight lines.

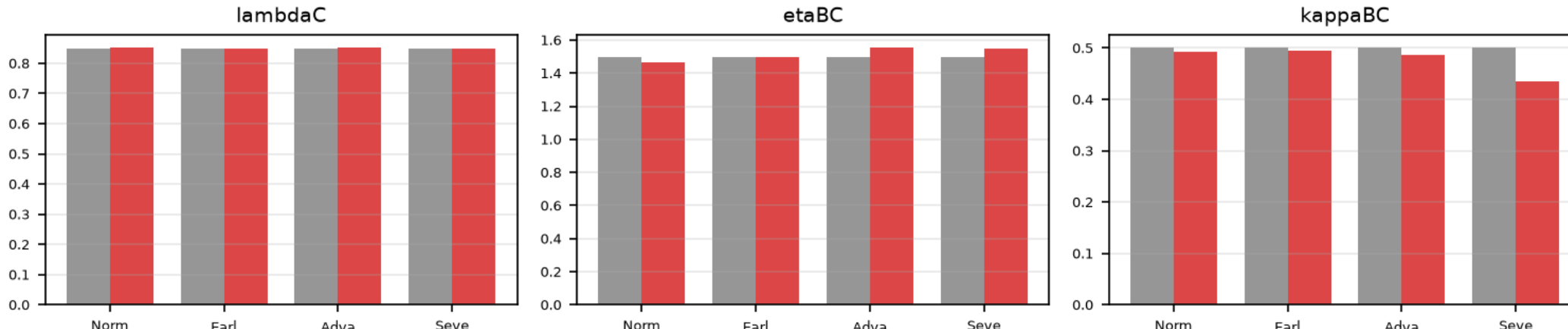


Trained on **40 sparse observations** plus the physics residual, the PINN tracks the Radau reference through the ATRA window in every regime.

1.06% **2.41%**

dense-label baseline, grand mean rel-L2 from 40 observations — the honest solve

Running it backwards



True (grey) against recovered (red), three of the best-converging parameters. Recovery is ill-posed at 36: the PINN-specific ceiling is 8/36, from gradient starvation and autodiff derivative bias.

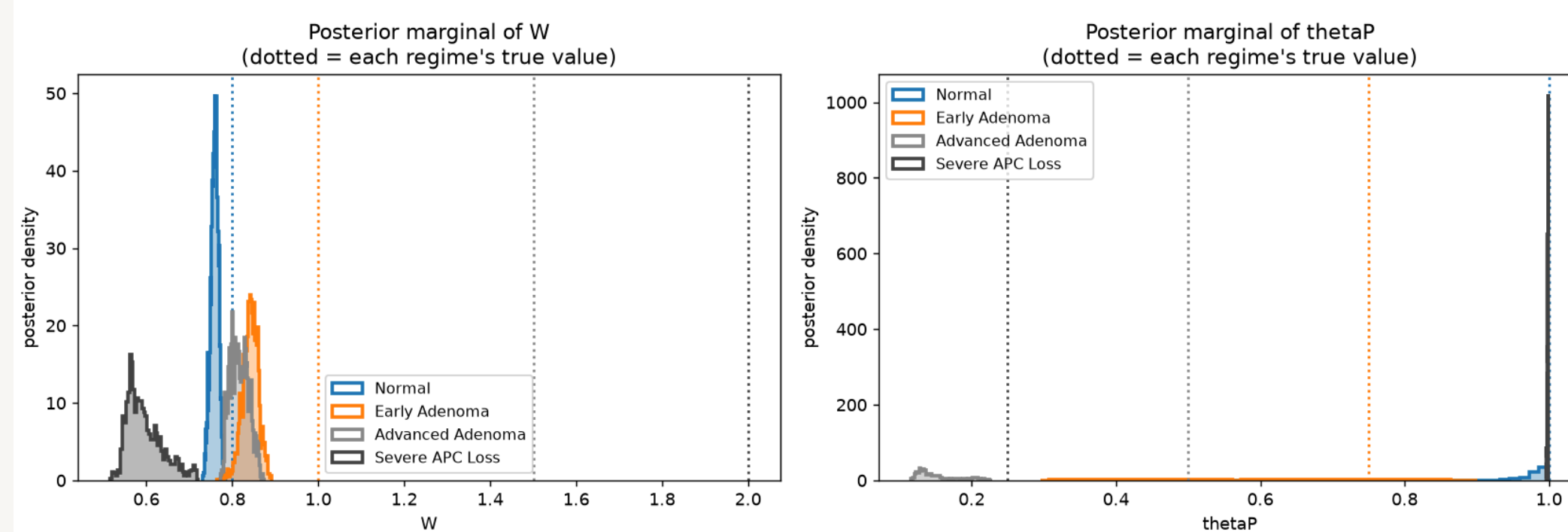
A derivative-free integral residual breaks half that ceiling

Enforcing the ODE in integrated form uses network **values** only, never the biased derivative. Recovery goes **10/4/5/7 $\rightarrow$ 17/16/10/7**, a same-conditions total of **37 $\rightarrow$ 50** of 144. Severe APC Loss does not move — what remains is structural, not numerical.

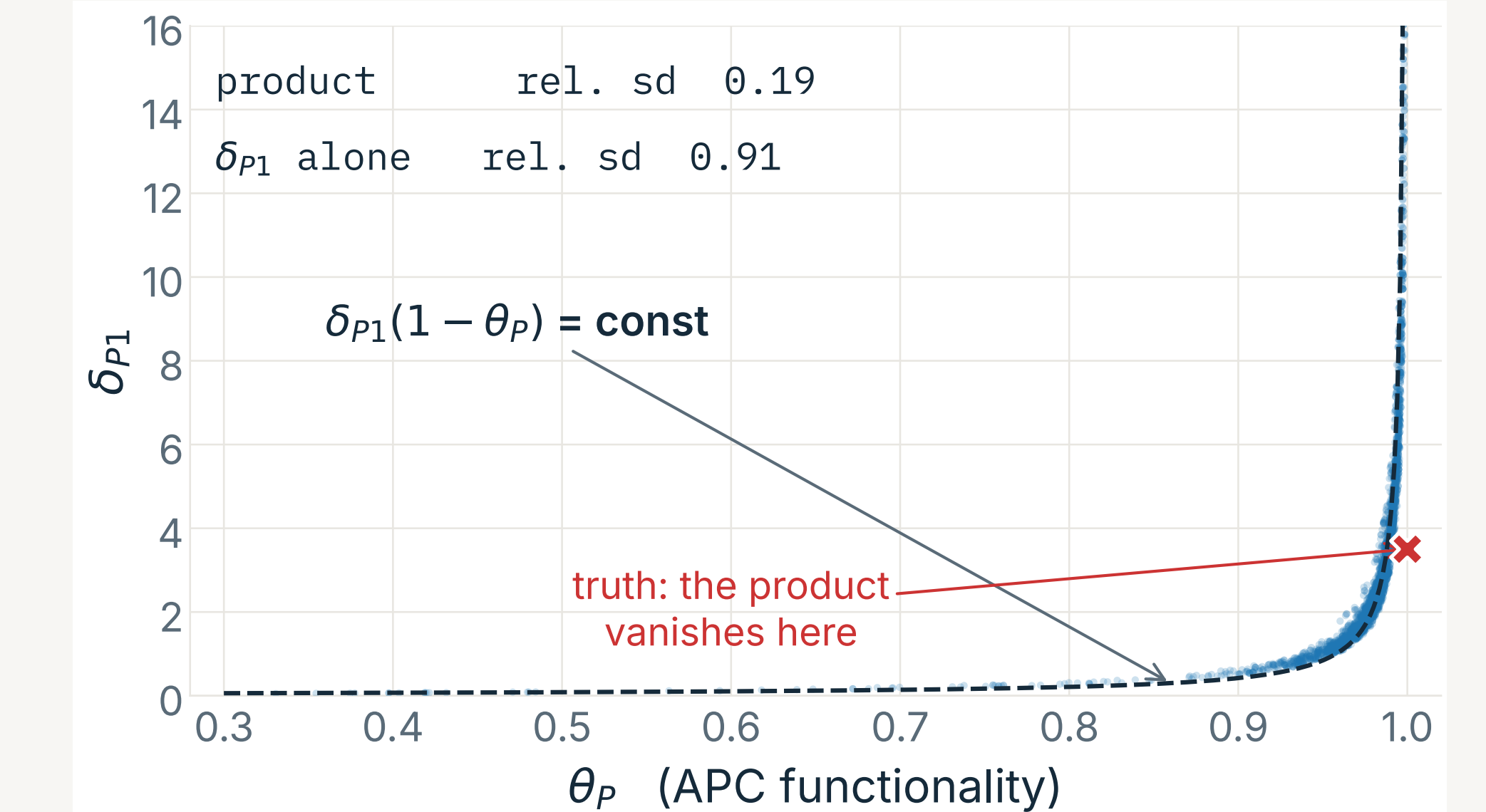
3 BAYESIAN INFERENCE

A posterior says what the data can never pin down.

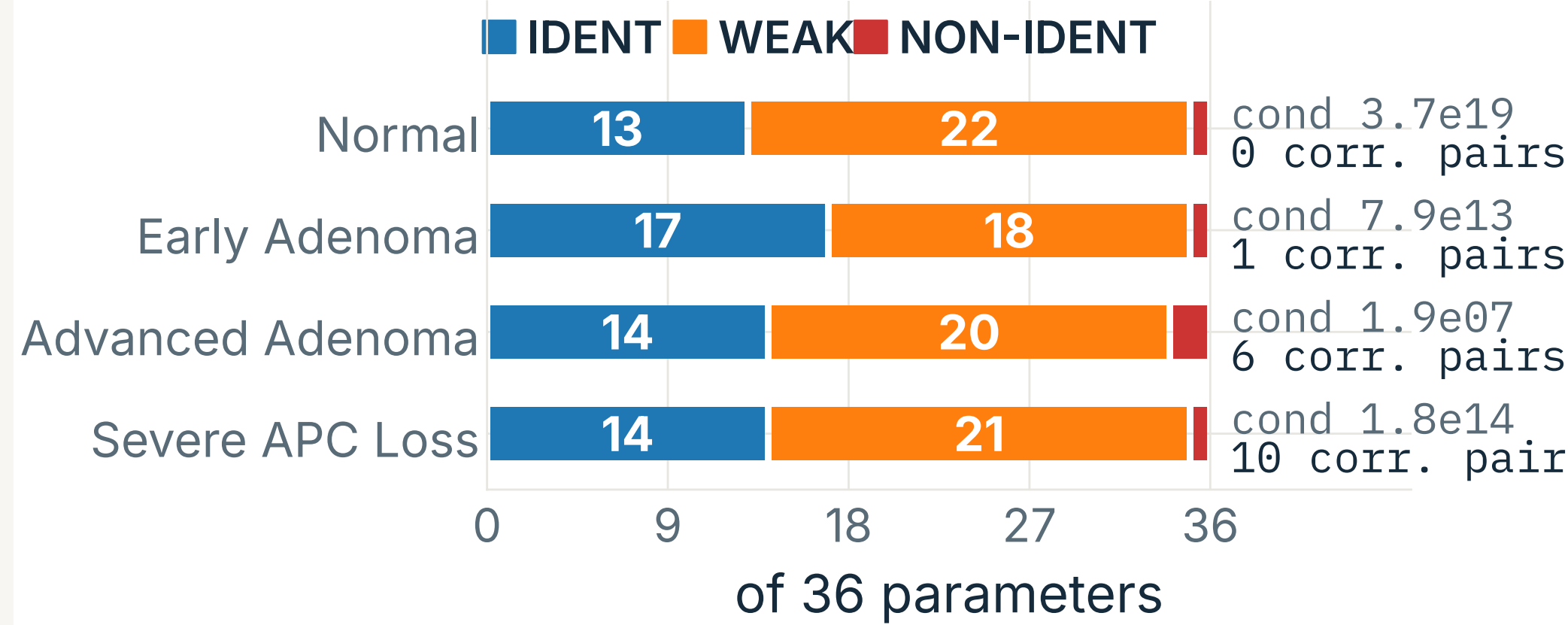
A point estimate cannot say **why** a parameter failed. We run HMC over the 36 parameters with the state networks frozen — the Bayesian twin of the point refine stage, and derivative-free, because the likelihood reuses the same integral residual.



W stays tight in every regime. θ_P reverts toward its prior as the WNT drive rises — the posterior-width statement of the high-WNT wall.



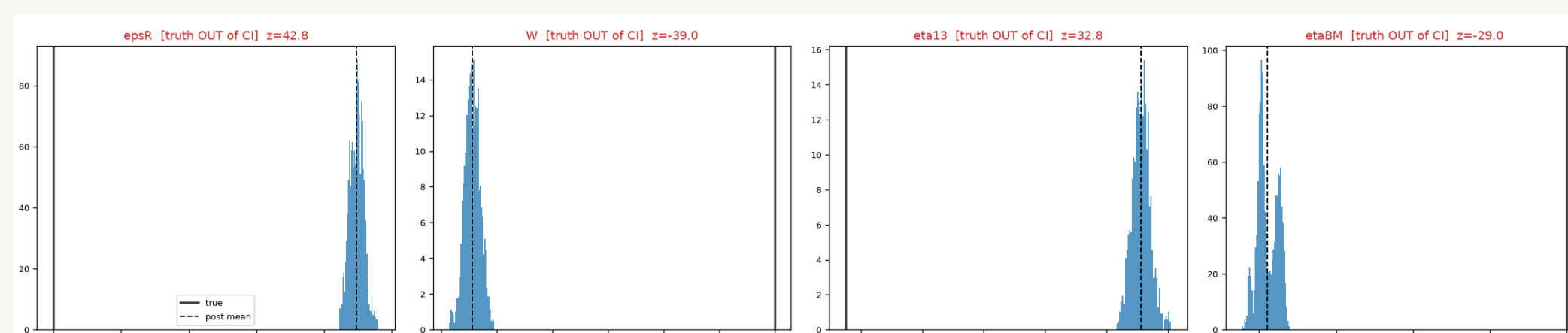
Unprompted, the sampler rides a curved valley: the **product** $\delta_{P1}(1 - \theta_P)$ is 4.7 $\times$ better constrained than δ_{P1} alone. At the healthy truth $\theta_P = 1$ the product vanishes — so δ_{P1} is structurally free.



The Fisher information agrees from a third direction: conditioning degrades with WNT drive, to 3.7×10^{19} in Severe APC Loss.

The first run said 36/36. It was an artifact.

The physics likelihood summed 210k residual terms against a tiny σ , producing a pinpoint — and biased — posterior. Below: truths far outside their own 95% intervals, coverage 13/36.



Honest recovery is **19/20/11/8**, not 36/36. The run still fails its own effective-sample-size and coverage gates, so we claim the **geometry** and the diagnosis — never the marginal widths.

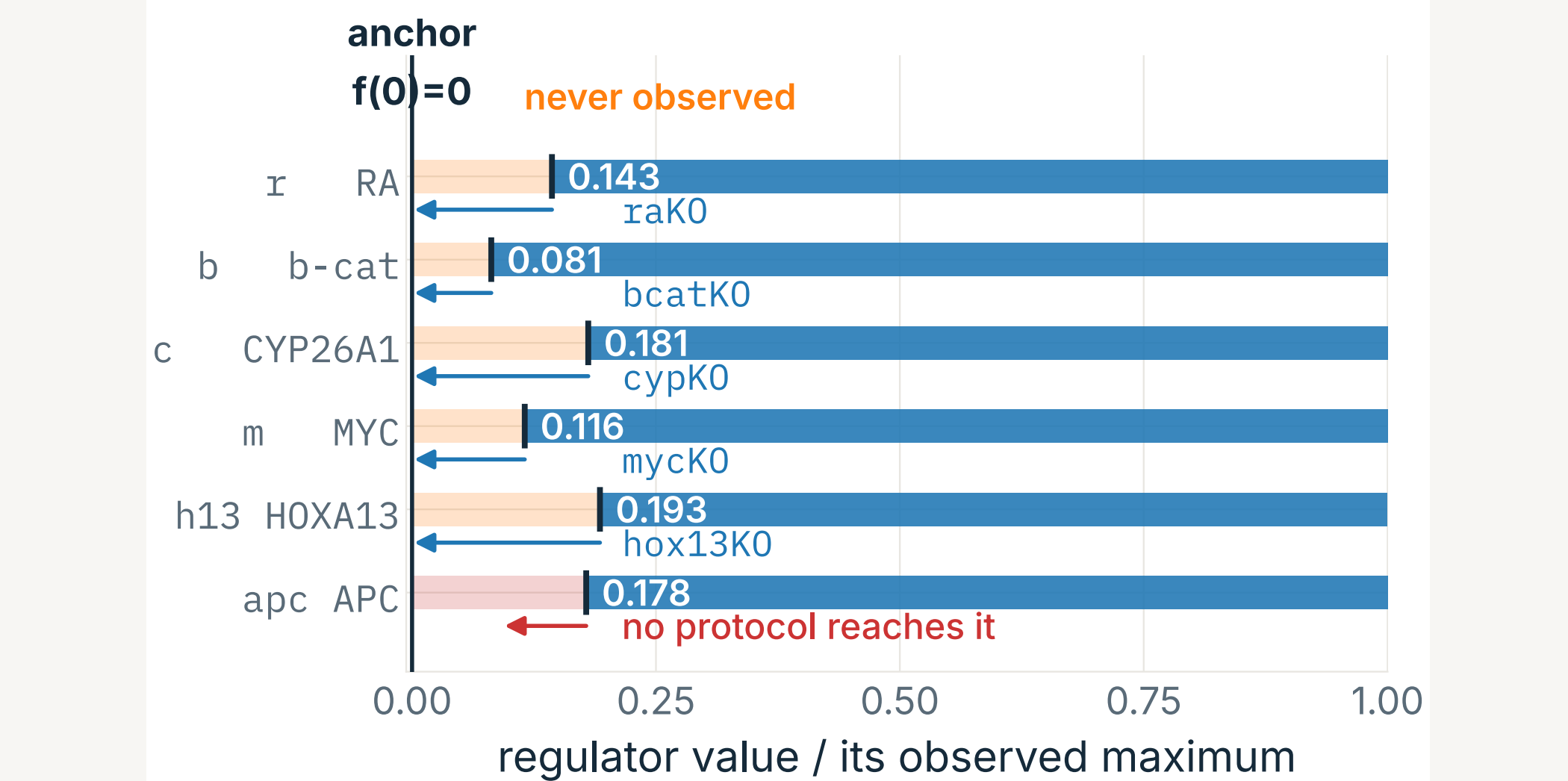
4 NEURAL-MECHANISTIC HYBRID

Replace a term with a network and identifiability returns.

$$\varepsilon \dot{x} = a + f_{\text{NN}}(u) - x$$

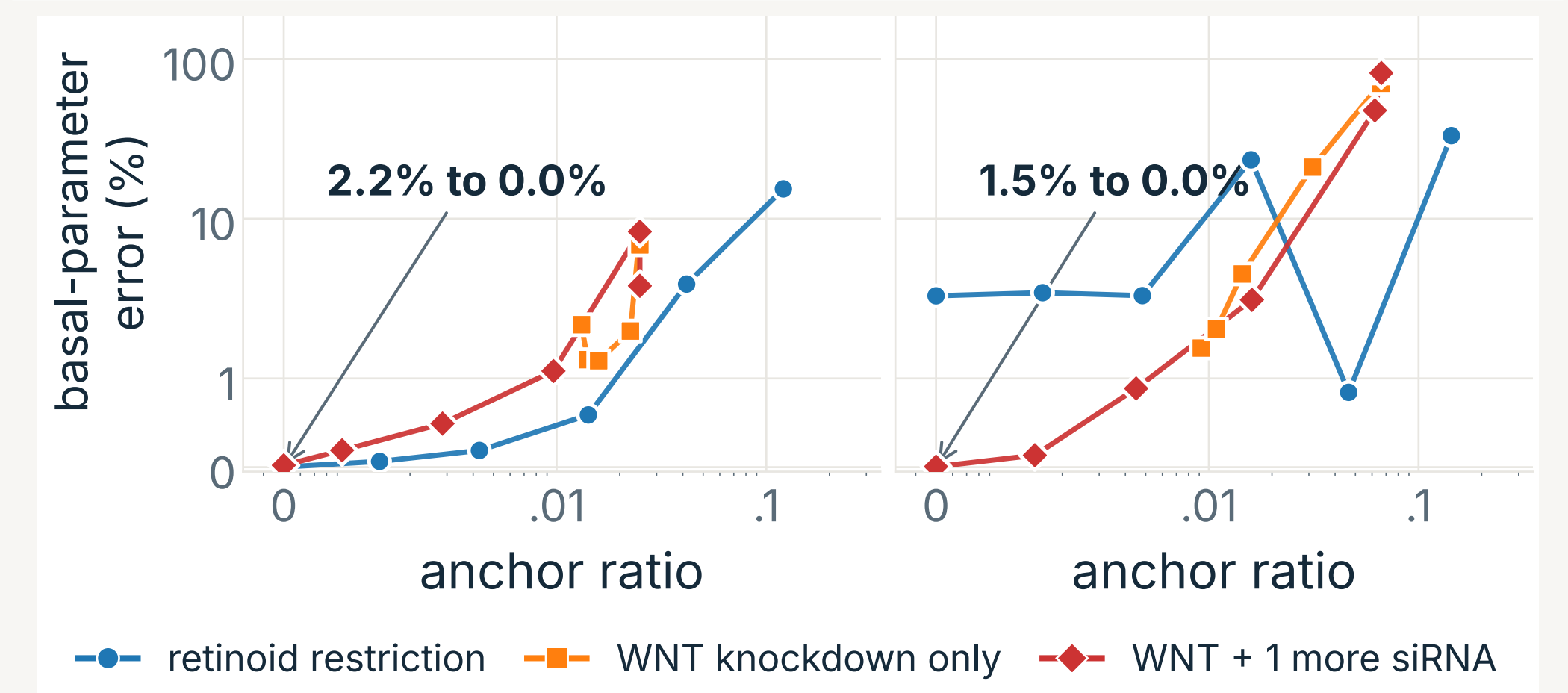
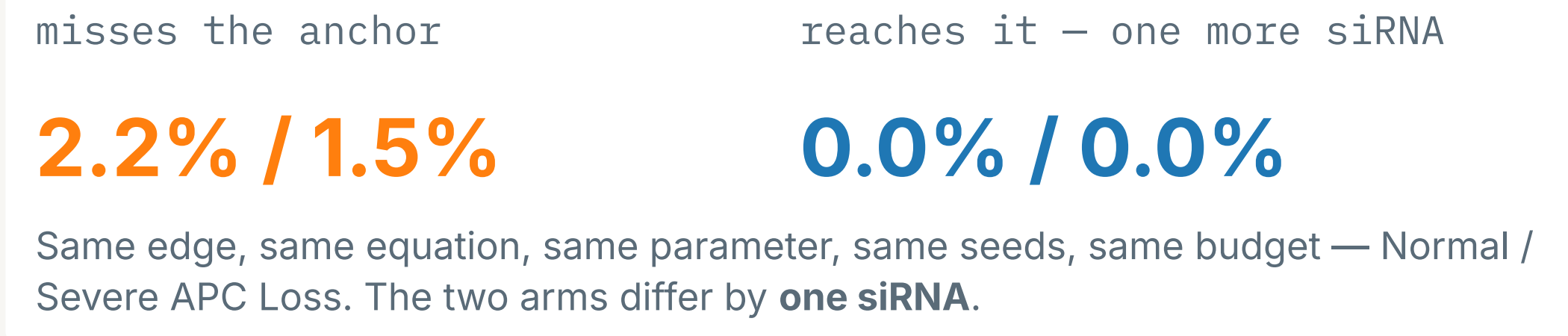
the anchor $f_{\text{NN}}(0) = 0$ is meant to stop the network absorbing a constant out of the basal-production parameter a

It does not work: across **five** parameterisations — including the monotone-and-bounded one the UDE literature recommends — the basal parameter comes out **14–203%** wrong.



Because across all 10 conditions and all 4 regimes, **no regulator ever approaches zero**. The anchor is a constraint asserted at a point the data never visit.

The fix is an experiment, not an architecture
A dose–response with the **condition count held fixed at eleven** at every dose — so “more data helps” is ruled out by construction.



Against the **anchor ratio** the arms collapse onto one curve; against dose they do not. The anchor, not the intervention, is the governing variable.

Takeaways

- 1 Nondimensionalization plus Fourier features makes a stiff, multiscale 7-ODE system solvable by a PINN from sparse data.
- 2 Inverse recovery is limited by **information**, not architecture — and a posterior says exactly which directions are unidentifiable.
- 3 A constraint on a learned term carries information only where the data have support; the remedy is experiment design, computable in advance.

What these numbers are not

Hybrid numbers come from an equation-local screen fed exact states — an upper bound, though both arms are fitted identically so the differences are fair. Restart noise ± 0.5 – 0.9 pp; single seed, so no error bars. Not shown: an information-matched control (no gain in 8 of 8 cells) and a prospective test ($95.6\% \rightarrow 0.0\%$). 4 of 6 pre-registered predictions failed and are reported as failures.